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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

PHARMACY MEDICINE STOCK AND BILLING MANAGEMENT SYSTEM

Submitted by

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Academic Year: **2025 – 2026**

APRIL 2026

Semester: **II**

DECLARATION

I hereby declare that the project entitled "**Pharmacy medicine stock and billing management system"** submitted in partial fulfillment of the requirements for the course **B.Sc Information Technology** is a **record of my original work** carried out under the guidance of **R Mugilan**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

Mithra M

CERTIFICATE

This is to certify that the project entitled **"Pharmacy medicine stock and billing management system"** is a **bona fide work** carried out by **Mithra M, B.Sc.Information Technology**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

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Head of the Department
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ABSTRACT

The Pharmacy Medicine Stock and Billing Management System is a Java-based application developed to streamline and automate the operations of medical stores and pharmacies. In many small and medium-sized pharmacies, stock management and billing are handled manually, which can lead to errors in inventory tracking, incorrect billing, and delays in customer service. This project aims to provide an efficient and reliable solution for managing medicine stock and generating accurate bills.

The system allows users to add, update, and manage medicine details such as name, quantity, price, expiry date, and supplier information. It also facilitates real-time stock monitoring, helping to prevent issues like overstocking or stock shortages. During billing, the user can select medicines, specify quantities, and the system automatically calculates the total amount, including applicable taxes and discounts.

Developed using Core Java concepts such as classes, objects, inheritance, exception handling, and file handling, the application ensures data accuracy and ease of use. It can be implemented as a console-based system or enhanced with a graphical user interface using Java Swing.

The main objective of this system is to improve efficiency, minimize human errors, maintain proper inventory records, and provide quick and accurate billing services. This project serves as a practical example of object-oriented programming and lays the groundwork for more advanced pharmacy management solutions.

1. INTRODUCTION

In the healthcare sector, efficient management of medicine stock and billing is essential for the smooth operation of pharmacies and medical stores. Traditionally, many small pharmacies rely on manual methods to maintain stock records and generate bills. These methods involve handwritten registers or basic calculations, which are time-consuming and prone to errors. Such inefficiencies can lead to incorrect billing, difficulty in tracking inventory, medicine shortages, or wastage due to expired products.

With the advancement of computer technology, automated pharmacy management systems have become increasingly important. A computerized system helps in maintaining accurate records of medicines, tracking stock levels in real-time, and generating bills quickly and efficiently. This not only improves operational efficiency but also enhances customer service by reducing waiting time.

The Pharmacy Medicine Stock and Billing Management System is a Java-based application developed to automate the process of inventory management and billing in pharmacies. The system allows users to store medicine details such as name, price, quantity, and expiry date, and manage stock efficiently. During billing, the user can select medicines, enter the required quantity, and the system automatically calculates the total cost along with applicable taxes and discounts.

The project is developed using Core Java programming concepts including classes, objects, methods, conditional statements, loops, and exception handling. It can be implemented as a console-based application or enhanced with a graphical user interface using Java Swing for better user interaction.

The primary objective of this system is to reduce manual workload, improve accuracy in stock management and billing, and provide faster and more reliable service to customers. This project also demonstrates the practical application of Java programming in developing real-world healthcare management solutions.

1.1 Objective

The main objective of the Pharmacy Medicine Stock and Billing Management System is to develop a simple and efficient application using Core Java that automates the process of medicine stock management and billing in pharmacies.

The specific objectives of this project are:

- To design and develop a Java-based pharmacy management system.
- To automate the calculation of medicine prices and total bill.
- To reduce manual errors in stock management and billing.
- To provide a user-friendly system for pharmacy staff to generate bills quickly.
- To maintain accurate records of medicine stock and expiry dates.
- To demonstrate the implementation of Core Java programming concepts such as classes,

objects, methods, loops, and conditional statements.
• To generate a clear and organized bill for customers.

1.2 Overview

The Pharmacy Medicine Stock and Billing Management System is a menu-driven application developed using the Java programming language. The system provides a digital platform where pharmacy staff can manage medicine inventory and generate bills automatically.

The system stores details of medicines such as name, price, quantity, and expiry date. The user can add new medicines, update stock, and view available inventory. During billing, the user selects the required medicines and enters the quantity purchased by the customer. The program calculates the total price for each medicine and generates a final bill including applicable taxes.

The system simplifies pharmacy operations by eliminating manual calculations and reducing the chances of errors in billing and stock tracking. The application can be developed as a console-based program or as a graphical user interface using Java Swing.

This project demonstrates the practical use of Java programming in developing real-world applications for healthcare and pharmacy management.

2. PROGRAM ANALYSIS

The Pharmacy Medicine Stock and Billing Management System is designed to overcome these issues by providing an automated solution for managing medicine inventory and generating bills. The system ensures accurate record-keeping of medicines, including their quantity, price, and expiry date, and helps in quick bill generation.

By analyzing the requirements, the system is designed to be user-friendly, efficient, and reliable, helping pharmacy staff perform daily operations smoothly while reducing manual effort and errors.

2.1 Problem Definition

In many small pharmacies, stock management and billing are often performed manually using registers or basic calculators. This traditional method creates several challenges.

Some common problems include:
• Errors in bill calculation
• Difficulty in tracking medicine stock
• Time-consuming billing process
• Lack of proper inventory records
• Risk of selling expired medicines

The Pharmacy Medicine Stock and Billing Management System is designed to solve these problems by providing a computerized solution that automates inventory management and billing processes.

2.2 Project Overview

The Pharmacy Medicine Stock and Billing Management System is a Java-based application developed to simplify stock management and billing in pharmacies.

The system provides a menu-driven interface where users can manage medicine details and generate bills efficiently. The user can add medicines, update stock, and select medicines based on customer requirements. The system automatically calculates the total price and generates a final bill.

The system includes the following functionalities:
• Displaying medicine details with prices
• Adding and updating medicine stock

• Selecting medicines and entering quantity
• Calculating total cost automatically
• Computing subtotal and tax amount
• Generating the final bill for the customer

This project is implemented using Core Java programming concepts and can be developed as either a console-based program or a Java Swing GUI application.

2.3 Scopes and Limitations

The scope of the Pharmacy Medicine Stock and Billing Management System includes the following:
• Managing medicine details including name, price, quantity, and expiry date
• Tracking and updating medicine stock in real-time
• Calculating the total bill automatically during billing
• Generating a detailed invoice for customers
• Demonstrating the practical use of Java programming concepts

The system can be used in:
• Pharmacies
• Medical stores
• Hospitals with in-house dispensaries
• Small clinics
• Online medicine shops

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The Pharmacy Medicine Stock and Billing Management System is designed as a menu-driven application where users can manage medicine stock, update quantities, and generate bills automatically.

3.1 Modules

The main modules of The Pharmacy Medicine Stock and Billing Management **System** are explained below:

1. Medicine Display Module

This module is responsible for displaying the list of available medicines along with their prices and stock quantity. The list helps the user select the required medicines for billing and stock management.

Example Medicine List:

1. Paracetamol 500mg ₹20 (50 tablets)
2. Amoxicillin 250mg ₹35 (30 capsules)
3. Cough Syrup 100ml ₹50 (20 bottles)
4. Vitamin D 1000IU ₹25 (40 tablets)
5. Antacid Tablets ₹15 (60 tablets)

The medicine list is displayed when the program starts so that the user can easily view available options and select medicines for billing or stock updates.

2.Medicine Selection Module

This module allows the user to select medicines from the available list and enter the quantity required by the customer.

Functions of this module include:

• Accepting user input for medicine selection
• Accepting the quantity of selected medicines
• Storing order details temporarily for billing

The system allows users to select multiple medicines before generating the final bill, ensuring accurate calculation and easy management of customer purchases.

3. Bill Calculation Module

This module performs all the billing calculations for the selected medicines.

The system calculates:
• Total price of each medicine (Price × Quantity)
• Subtotal of all selected medicines
• Applicable tax (GST)
• Final payable amount

All calculations are performed automatically using Java programming logic.

Example Calculation:

- Item Total = Price × Quantity
- GST = 5% of Subtotal
- Final Bill = Subtotal + GST

This module ensures accurate and fast billing for pharmacy customers.

4.Invoice Generation Module

This module generates the final bill for the customer after medicine selection and billing calculations.

The invoice includes:
• Medicine name
• Quantity purchased
• Price per unit
• Subtotal
• GST amount
• Final bill amount

Example Bill Output:

------ PHARMACY BILL ------

Medicine Name	**Qty**	**Price**
Paracetamol 500mg	**2**	**40**
Cough Syrup 100ml	**1**	**50**
Subtotal		**90**
GST (5%)		**4.5**

Total **94.5**

Thank You! Get Well Soon!

5. Exit Module

This module allows the user to exit the application safely after completing the billing process.

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the Pharmacy Medicine Stock and Billing Management System, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of pharmacy operations.

The main problems identified in manual pharmacy systems are:

- Manual calculations may lead to billing errors.
- Managing medicine stock manually is time-consuming.
- Difficult to track expiry dates and stock levels efficiently.
- Lack of automated records for transactions and inventory.

Based on these problems, the system requirements were defined to develop an automated pharmacy management application that simplifies stock management and billing operations.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

• Medicine Display Module – Shows available medicines with price and stock details.

• Medicine Selection Module – Allows users to select medicines and enter quantities.

• Bill Calculation Module – Calculates total price, subtotal, and applicable taxes.

• Invoice Generation Module – Generates a detailed bill for the customer.

• Exit Module – Safely closes the application after completing operations.

The program flow and structure are designed so that pharmacy staff can easily interact with the system, manage stock efficiently, and generate accurate bills.

4.3. Program Development

After designing the system, the program is implemented using the Java programming language.

The following steps were followed during development:

• Creation of classes and methods for managing medicines and billing

• Implementation of menu-driven logic for user interaction

• Writing code for medicine selection and bill calculation

• Handling user input for stock management and billing

• Displaying the final bill and generating invoices

Core Java concepts such as classes, objects, loops, conditional statements, and methods were used to develop an efficient and user-friendly pharmacy management system.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	Expected Result
Select medicine	Medicine displayed correctly
Enter quantity	Correct quantity accepted
Calculate bill	Correct total displayed
Invalid input	Error handled properly

This phase ensures that the Pharmacy Medicine Stock and Billing Management System runs without errors, handles exceptions, and produces accurate billing and stock management results.

4.5. Result and Documentation

After successful testing, the results of the system are documented. The system generates the final bill based on the selected medicines and their quantities.

The results are demonstrated through program execution and screenshots showing the input and output of the system.

Proper documentation is prepared to explain the design, implementation, and working of the Pharmacy Medicine Stock and Billing Management System. This includes details of modules, program flow, test cases, and sample invoices for reference.

4.6. Algorithm

Algorithm for the Pharmacy Medicine Stock and Billing Management System:

1. Start the program.
2. Display the list of available medicines with prices and stock quantity.
3. Ask the user to select a medicine.
4. Enter the quantity of the selected medicine.
5. Calculate the total cost of the selected medicine.
6. Add the cost to the total bill.
7. Repeat steps 3–6 for additional medicines if required.
8. Calculate the GST or applicable tax amount.
9. Calculate the final bill amount.
10. Display the final bill to the user.
11. End the program.

4.7. Flowchart

Students should draw a flowchart showing program flow.

Example flow:

Start
↓
Display Medicine List
↓
Select Medicine
↓
Enter Quantity
↓
Calculate Bill
↓
Generate Invoice

↓
End

Students can draw using:

- draw.io
- Lucidchart
- MS Visio

5. SOURCE CODE

```
import java.util.*;

class Medicine {
    String name;
    double price;
    int quantity;

    Medicine(String name, double price, int quantity) {
        this.name = name;
        this.price = price;
        this.quantity = quantity;
    }
}

public class PharmacyBillingSystem {
    static Scanner sc = new Scanner(System.in);
    static ArrayList<Medicine> stock = new ArrayList<>();
    static ArrayList<Medicine> cart = new ArrayList<>();
```

```
public static void main(String[] args) {
    initializeStock();
    boolean running = true;
    while (running) {
        System.out.println("\n===== PHARMACY BILLING SYSTEM =====");
        System.out.println("1. Display Medicines");
        System.out.println("2. Purchase Medicine");
        System.out.println("3. Generate Bill");
        System.out.println("4. Exit");
        System.out.print("Enter your choice: ");
        int choice = sc.nextInt();
        switch (choice) {
            case 1 -> displayMedicines();
            case 2 -> purchaseMedicine();
            case 3 -> generateBill();
            case 4 -> {
                System.out.println("Exiting... Thank You!");
                running = false;
            }
            default -> System.out.println("Invalid choice! Try again.");
        }
    }
}

static void initializeStock() {
    stock.add(new Medicine("Paracetamol 500mg", 20, 50));
    stock.add(new Medicine("Amoxicillin 250mg", 35, 30));
    stock.add(new Medicine("Cough Syrup 100ml", 50, 20));
    stock.add(new Medicine("Vitamin D 1000IU", 25, 40));
```

```
    stock.add(new Medicine("Antacid Tablets", 15, 60));
}

static void displayMedicines() {
    System.out.println("\nAvailable Medicines:");
    System.out.println("S.No\tMedicine\t\tPrice\tStock");
    int i = 1;
    for (Medicine m : stock) {
        System.out.println(i + "\t" + m.name + "\t" + m.price + "\t" + m.quantity);
        i++;
    }
}

static void purchaseMedicine() {
    displayMedicines();
    System.out.print("Enter medicine number to purchase: ");
    int choice = sc.nextInt();
    if (choice < 1 || choice > stock.size()) {
        System.out.println("Invalid selection!");
        return;
    }
    Medicine selected = stock.get(choice - 1);
    System.out.print("Enter quantity: ");
    int qty = sc.nextInt();
    if (qty > selected.quantity) {
        System.out.println("Insufficient stock!");
        return;
    }
    cart.add(new Medicine(selected.name, selected.price, qty));
```

```
            selected.quantity -= qty;
            System.out.println("Added to cart!");
        }

        static void generateBill() {
            if (cart.isEmpty()) {
                System.out.println("No medicines purchased yet.");
                return;
            }
            double subtotal = 0;
            System.out.println("\n------ PHARMACY BILL ------");
            System.out.println("Medicine\tQty\tPrice\tTotal");
            for (Medicine m : cart) {
                double total = m.price * m.quantity;
                subtotal += total;
                System.out.println(m.name + "\t" + m.quantity + "\t" + m.price + "\t" + total);
            }
            double gst = subtotal * 0.05; // 5% GST
            double finalBill = subtotal + gst;
            System.out.println("\nSubtotal: " + subtotal);
            System.out.println("GST (5%): " + gst);
            System.out.println("Total Bill: " + finalBill);
            System.out.println("Thank You! Get Well Soon!");
            cart.clear(); // Clear cart after billing
        }
    }
```

6.INPUT AND OUTPUT SCREENS

```
Microsoft Windows [Version 10.0.26200.8039]
(c) Microsoft Corporation. All rights reserved.

D:\java>javac PharmacyBillingSystem.java

D:\java>java PharmacyBillingSystem

===== PHARMACY BILLING SYSTEM =====
1. Display Medicines
2. Purchase Medicine
3. Generate Bill
4. Exit
Enter your choice: 1

Available Medicines:
S.No    Medicine                Price   Stock
1       Paracetamol 500mg       20.0    50
2       Amoxicillin 250mg       35.0    30
3       Cough Syrup 100ml       50.0    20
4       Vitamin D 1000IU        25.0    40
5       Antacid Tablets 15.0    60

===== PHARMACY BILLING SYSTEM =====
1. Display Medicines
2. Purchase Medicine
3. Generate Bill
4. Exit
Enter your choice: 2

Available Medicines:
S.No    Medicine                Price   Stock
1       Paracetamol 500mg       20.0    50
2       Amoxicillin 250mg       35.0    30
3       Cough Syrup 100ml       50.0    20
4       Vitamin D 1000IU        25.0    40
5       Antacid Tablets 15.0    60
Enter medicine number to purchase: 4
Enter quantity: 7
Added to cart!

===== PHARMACY BILLING SYSTEM =====
1. Display Medicines
2. Purchase Medicine
3. Generate Bill
4. Exit
Enter your choice: 3

------ PHARMACY BILL ------
Medicine        Qty     Price   Total
Vitamin D 1000IU        7       25.0    175.0

Subtotal: 175.0
GST (5%): 8.75
Total Bill: 183.75
Thank You! Get Well Soon!

===== PHARMACY BILLING SYSTEM =====
1. Display Medicines
2. Purchase Medicine
3. Generate Bill
4. Exit
Enter your choice: 4
Exiting... Thank You!
```

7.

Discussion and Conclusion

The Pharmacy Medicine Stock and Billing Management System was developed to automate medicine stock management and billing in pharmacies. The system

allows users to view medicines, enter quantities, and generate bills automatically.

During the development process, Core Java programming concepts such as loops, conditional statements, methods, and object-oriented principles were applied. The system successfully manages stock updates, calculates bills, and generates accurate invoices.

The system was tested with different inputs to ensure correct stock management and billing. The results show that the system reduces manual effort, minimizes errors, and improves efficiency in pharmacy operations.

The Pharmacy Medicine Stock and Billing Management System successfully demonstrates the practical application of Core Java programming concepts in developing a real-world business application.

The system automates stock management and billing, reduces calculation errors, and provides faster service to customers. It simplifies pharmacy operations by generating accurate invoices and keeping track of medicine inventory.

This project helps students understand how Java programming can be used to develop practical software systems. Future enhancements could include database integration, a graphical user interface (GUI), expiry date tracking, and online payment features for a more advanced pharmacy management solution

8.REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

• https://www.oracle.com/java
• https://www.geeksforgeeks.org
• https://www.w3schools.com
• https://www.javatpoint.com